the safe end. Then the flared end was hammered down on the tube.

After the tubes were all pieced, the fire was prepared for the welding in the following way: Taking three fire bricks, we placed one on each side of the forge fire. The third brick was laid across the top and the fire was banked around some distance away.

When all was ready, with the blacksmith as a helper, we made the welds, placing the tube in the fire and constantly turning it until it became hot enough. It was tapped lightly with a long-handled light hammer, the top brick being removed for this operation. After this, it was allowed to heat again, then it was removed from the fire, quickly slipped over the mandrel, the helper striking the swedge lightly and quickly, the tube being turned on the mandrel until the weld was made. Then the tube was placed in the bottom swedge and rounded up.

After the welding and when the tubes had cooled off, they were tested for leaks by driving a wooden plug in the new end and then pouring water into the tube. If a leak developed, the tube was rewelded. A very small percent showed any signs of leakage.

The boiler now was measured for length of tubes, the tubes heated and cut to length with the buck saw.

Bay City, Mich. JOHN A. ANDERSON.

Concerning Staybolts

TO THE EDITOR:

A few short paragraphs about flexible staybolts in the February issue of THE BOILER MAKER, certainly starts a man to thinking! What are "flexible" staybolts, anyway? Are there any staybolts which are not flexible to a greater or less extent? Quite likely, if there are any non-flexible staybolts, they are the ones which break. Possibly, the staybolts which leak, are not flexible enough to meet the go and come of breathing movements of the sheets to which the staybolts are attached.

It does not require much study to enable a man to make up his mind that the staybolt has a deuce of a hard time—all the time—and that the wonder is, not why staybolts break, but how can any staybolts stand up under the stresses to which they are subjected.

Even with staybolts in their most favorable positions, say in plain inner and outer cylinders, it is doubted if there be one staybolt in the entire boiler, which is subjected merely to a straight tensile pull, due to the pressure upon the inner and outer cylinders.

Due to the irregular movements of portions of the inner and outer cylinders while pressure is being applied, it is evident to the thinking man that there will be developed sidewise pulls or stresses which tend to push one or both ends of a staybolt out of alinement. This sidewise movement of one end of a staybolt can only result in bending the staybolt unless it is sufficiently flexible to spring back to its original position after the side stresses have been removed.

Let's see what can and does happen when the inner and outer sheets of a boiler are not symmetrical, or are tied together in an irregular manner by gusset sheets, flanges, mud rings, fire doors or other parts securely riveted to the two cylinders to which the staybolts are securely attached.

How would you like to make a few simple tests which will show exactly in which way a staybolt in a boiler is being bent, or at least pushed by the stresses developed by the comparative movement of one stayed surface or the other? The following paragraphs show how to make individual tests of each staybolt, in a manner which allows its deviation, if any, to be observed and comparatively measured.

If possible, select for the tests a boiler fitted with hollow staybolts. If it be found necessary to use a vessel with undrilled staybolts, it will be necessary to drill the end of each staybolt to be tested. The holes need be drilled only about ½-inch deeper than the inner side of the shell plate and preferably, a drill should be used which will make a hole the same diameter as the holes in drilled staybolts. Next, procure two small reamers, one just large enough in diameter to clean out the holes in staybolts to be tested. This reamer should be run into each staybolt hole to a distance of at least ½ inch below the inner surface of the boiler sheet. The other reamer should be about 1/100 inch larger than the first mentioned reamer and this should be run into each staybolt to be tested, but only deep enough to barely reach flush with the inner side of the stayed sheet.

Next, procure some pieces of cold-rolled steel rod, just large enough to make a driving fit into the staybolt holes which have been reamed ½ inch below the inner surface of the boiler sheet. These pieces of cold-rolled steel should be about 30 inches long, provided there is room for them to project that distance when driven into the holes in staybolts. If necessary, shorter rods may be used, but, bear in mind that the longer these rods, the more apparent becomes the bending effect in any staybolt. The number of rods required, will be the number of staybolts which it is desired to test at the same time—a separate rod for each staybolt, will be required; but any number of staybolts can be tested with a single rod by making the tests one after the other. However, it is better to test several bolts at the same time, as the tests reveal also, the direction in which a staybolt is bent or flexed.

Procure also a similar number of pieces of cold-rolled steel of slightly larger diameter. In fact, this second batch of rods should be just large enough to be driven tightly into the larger holes, reamed plate-thickness deep, in the ends of the staybolts. These larger rods should be about 8 inches longer than the 30-inch rods.

To make a staybolt test, drive one of the smaller rods tightly into the staybolt to be tested, making sure that the rod does not touch the staybolt until below the inside of the shell plate. Into an adjoining staybolt drive one of the larger rods, taking care that the rod goes no deeper than flush with the inside of the shell plate. Now, bend the end of the longer rod until its tip lies directly over and in line with the 30-inch rod in the staybolt under test. The rods should not touch, but their ends should be fairly in line with each other. Fit up as many other staybolt test rods as you desire—the more the better—and then you can see how some staybolts are deflected in one direction and some in another.

With the test rods solidly in place, put the boiler under pressure, either by getting up steam or hydrostatically, and then watch the test rods. They will tell you how the staybolts are being affected by distortion stresses caused by shell movements.

Indianapolis, Ind. JAMES F. HOBART.

MULTIPLE RETORT STOKERS.—A new catalogue describing the CE multiple retort stoker has been published by the Combustion Engineering Corporation, 200 Madison Avenue, New York. This super-station type of stoker is a comparatively new design and is the result of three years' extensive research and development.

Questions and Answers

Problems in design, construction and repair of boilers, heavy plate and tank work

Conducted by George M. Davies

This department is open to subscribers of THE BOILER MAKER *for the purpose of helping those who desire assistance on practical boiler shop problems. All questions should be definitely stated and clearly written in ink, or typewritten, on one side of the paper and sketches furnished if necessary. Inquiries should bear the name and address of the writer. Anonymous communications will not be considered. The identity of the writer, however, will not be disclosed unless the editor is given permission to do so.*

Leaky Boiler Check

Q.—If boiler check leaks, what effect will it have on the injector and injector tubes? E. L. B.

A.—A leaking boiler check could cause the injector to fail to prime.

Leaking Injector Steam Ram

Q.—If the steam ram to injector leaks, what effect will it have on the injector? Will it cause injector tubes to corrode? E. L. B.

A.—The steam pipe to the injector leaking would have no effect on the working of the injector as long as boiler pressure could be maintained in the pipe.

Lift of Boiler Check

Q.—If a boiler check has too much lift, what effect will it have on the injector? E. L. B.

A.—The boiler check having too much lift would not necessarily affect the injector; however this condition would be injurious to the check valve seat.

Coated Injector Tubes

Q.—If boiler check leaks, or steam ram to injector leaks, will that cause injector tubes to coat over with scale or plug up the small holes in injector tube parts? E. L. B.

A.—The boiler check or injector steam pipe leaking should not cause the tubes to coat over with scale, or plug. This condition is no doubt caused by impure water.

Sediment in Branch Pipe

Q.—A boiler check has been leaking for five days and we have been unable to use the injector. (Lifting injector No. 8 or No. 9.) At wash out, the branch pipe was taken down and found full of soft sediment. What was the cause of so much sediment in branch pipe? E. L. B.

A.—The sediment in the branch pipe is no doubt due to the use of impure water. The impurities in the water not being able to pass through the boiler check valve.

Layout of Cone with Offet Cylinder

Q.—Enclosed please find a section of a cone with a cylinder at the top 6 inches off center. Will you please show me the proper way to develop the patterns for both the cone and the cylinder? I experimented with one of these a short time ago with not very good results. C. C. K.

A.—To develop the patterns for the intersection of a cone and cylinder as outlined in the question, draw the elevation, end view and plan as shown in Figs. 1, 2 and 3.

To develop the intersection line of the cylinder and cone in the elevation and end views, divide the plan of the cone *ABC* into any number of equal parts, 12 being taken in this case; the greater the number of equal parts taken the more accurate the final development. Number the divisions from *1* to *13* as shown; connect these points with the center *H* and where these lines cut the plan of the cylinder *DEF*, number these points from *1'* to *13'* as shown.

Perpendicular to the line *A-B* of the plan and passing through the points *1* to *13*, draw lines cutting the line *A'-B'* of the elevation number these intersections from *1* to *13* as shown and connect these points to *H'* of the elevation.

Perpendicular to the line *A-B* of the plan and passing through the point *1'* of the plan, draw a line cutting the line *H'-1* of the elevation locating the point *1'* of the elevation.

Perpendicular to the line *A-B* of the plan and passing through the point *2'* of the plan, draw a line cutting the line *H'-2* of the elevation, locating the point *2'* of the elevation.

In the same manner draw lines perpendicular to *A-B* through the points *3'* to *13'* of the plan cutting the corresponding lines *H'-3'* to *H'-13'* of the elevation, locating the points *3'* to *13'* of the elevation. Connect the points *1'* to *13'* of the elevation, thus completing the intersection line of the cylinder and cone in the elevation.

The end view can now be constructed by projection, all the points having been located in either the plan or elevation.

The next step in the development of the cone is to construct a series of right-angle triangles to obtain the true length of the surface lines of the cone as *1-1'*, *2-2'*, *3-3'*, etc. of the elevation.

Draw the line *M-N*, Fig. 4, and at *O* erect a perpendicular *O-R* to *M-N*. With *O* as a center and the dividers set equal to the distance *2-2'* of the plan, scribe an arc cutting the line *M-N* at *2*, then with *O* as a center and the dividers set equal to the perpendicular distance between the line *A'-B'* and the point *2'* of the elevation, scribe an arc cutting the perpendicular line *O-R* at *2'*. Connect the points *2* and *2'*, Fig. 4. This line is then the true length of the surface line *2-2'* of the elevation.

With *O* as a center and the divider set equal to the distance *3-3'* of the plan, scribe an arc cutting the line